

INC-0087

The Storage Nobody Can Access · RBAC · Private Endpoint · DNS · Storage Firewall

3

Problems Found

3

Problems Fixed

0

Portal Clicks

100%

CLI / PowerShell

Engineer

Alan — Azure Security Engineer

Ticket**INC-0087 · CRITICAL Priority****Reported By**

Marcus T. — DevOps Lead

Environment

Azure · canadacentral · lab2-rg

Tools Used

PowerShell · Azure CLI · Bicep

Cert Alignment

AZ-104 + AZ-500

□ The Ticket

"Alan, we have a storage account deployed yesterday and nothing can reach it. Three separate problems: (1) I have a role assignment but I get 403 AuthorizationFailure when uploading a blob. (2) Our app inside the VNet tries to connect via the private endpoint but keeps getting the public IP instead of a private one. (3) Even after fixing the DNS, the storage firewall is blocking traffic. Production is at risk." — Marcus T., DevOps Lead

The infrastructure was already deployed. Three separate failures, all presenting as "access denied" but each for a completely different reason. This is what makes storage troubleshooting hard — the symptoms look the same on the surface.

What I Found & Fixed

PROBLEM 1 Wrong RBAC Role

SYMPTOM

403 AuthorizationFailure when uploading a blob — despite having a role assignment on the storage account.

ROOT CAUSE

The assigned role was "Storage Account Contributor" — this lets you manage account settings but gives ZERO access to blob data. It is like having a key to the building but not to the filing cabinet inside. The role needed was "Storage Blob Data Contributor" which grants actual read/write access to blob contents.

FIX

Removed Storage Account Contributor. Assigned Storage Blob Data Contributor to the correct principal.

PROBLEM 2 Private Endpoint DNS Broken

SYMPTOM

App inside VNet resolves storage hostname to public IP (20.x.x.x) instead of private IP (10.1.2.4). Traffic leaving the VNet to reach storage — defeating the entire purpose of a private endpoint.

ROOT CAUSE

Two components were missing: (1) The Private DNS Zone was never linked to the VNet — so VMs inside the VNet could not use the private directory. (2) No DNS Zone Group existed on the private endpoint — so the private IP was never auto-registered in the DNS zone. Without both, Azure falls back to public DNS.

FIX

Created VNet link on the Private DNS Zone. Created DNS Zone Group on the private endpoint. Azure automatically registered the A record: lab2sace7exhub5u5j6 → 10.1.2.4.

PROBLEM 3 🔥 Storage Firewall Blocking VNet Traffic

● SYMPTOM

Even with correct DNS resolution, traffic from the VNet subnet was rejected by the storage account firewall.

🔍 ROOT CAUSE

Storage firewall defaultAction was set to Deny with an empty virtualNetworkRules list. Additionally, the pe-subnet had no Microsoft.Storage Service Endpoint configured — Azure requires this before a subnet can be added to storage firewall rules.

✅ FIX

Enabled Microsoft.Storage Service Endpoint on lab2-pe-subnet. Added the subnet to the storage account network rules. Traffic from the VNet now reaches storage through the approved path.

✓ Verification

#	What Was Verified	Command	Result
1	Role assignment updated to correct role	<code>az role assignment list --scope ...</code>	✓Storage Blob Data Contributor
2	DNS Zone linked to VNet	<code>az network private-dns link vnet list</code>	✓Completed
3	DNS Zone Group on private endpoint	<code>az network private-endpoint dns-zone-group list</code>	✓10.1.2.4 registered
4	VNet subnet in storage firewall rules	<code>az storage account show --query networkRuleSet</code>	✓lab2-pe-subnet allowed
5	End-to-end blob upload success	<code>az storage blob upload --auth-mode login</code>	✓Finished 100%

These are the engineering principles applied in this lab — not just commands, but real decision-making under pressure.

RBAC is not just about having a role — it is about having the RIGHT role

Storage Account Contributor and Storage Blob Data Contributor are completely different permissions. One manages the account. The other accesses the data. Confusing them is one of the most common productions RBAC mistakes in Azure.

Private endpoint DNS requires TWO components — miss either one and it breaks

A VNet Link tells the VNet which DNS zone to use. A DNS Zone Group auto-registers the private IP. Both must exist. Either missing = DNS falls back to public internet = private endpoint is useless.

Storage firewall Deny means Deny everything — subnets need Service Endpoints first

Before adding a subnet to storage firewall rules, Microsoft.Storage Service Endpoint must be enabled on that subnet. This is a prerequisite that is easy to miss and causes confusing errors.

Troubleshoot in layers — RBAC first, then DNS, then firewall

Each layer depends on the one below it. Fixing firewall before DNS is fixed is pointless. Fixing DNS before RBAC is fixed changes nothing. Order matters.